The two-ρ terms of the three-ρ equation

These are the two-ρ terms obtained from the final three-ρ equation (`ThreeRho_final.tex`, black and blue lines). They come row by row, in the order of that equation.

- **How they are obtained.** Every line is reordered in the charge order written in the three-ρ equation, with $[\rho^a(x),\rho^b(y)] = if^{abc}\rho^c(x)\,\delta^{(2)}(x-y)$:

$$\begin{aligned} ABC &= S(ABC) + \tfrac{1}{4}\big(\{[A,B],C\} + \{[A,C],B\} + \{[B,C],A\}\big),\\ A\{B,C\} &= 2S(ABC) + \tfrac{1}{2}\{[A,B],C\} + \tfrac{1}{2}\{[A,C],B\},\\ \{B,C\}A &= 2S(ABC) - \tfrac{1}{2}\{[A,B],C\} - \tfrac{1}{2}\{[A,C],B\}. \end{aligned}$$

  The symmetrized parts $S(ABC)$ cancel (three-ρ equation), and one-ρ terms are dropped. The δ-functions are carried out and the colour algebra is done with $U^{bc} = U^{cb}$, $U^{ab}(p)U^{ac}(p) = \delta^{bc}$, $f^{abc}U^{ad}(p)U^{be}(p)U^{cg}(p) = f^{deg}$ and $f^{acd}f^{bcd} = N_c\delta^{ab}$. Terms with the same kernel and the same colour structure are added within each row. The c.c. is included.

- **Variables.**
  - The measured gluon is at w (amplitude) and w' (c.c.); the unobserved gluon is at z.
  - The two remaining charges are x and x'. The charge whose emission kernel ends at w is x, and the one ending at w' is x'. If one charge emits at both w and w', it is called x' and the other one x.
  - $\mathcal{K}(X) = X/X^2$.

- **Charges.** The two charges form a symmetric product: $\rho^a(p)\,\rho^b(q) \equiv \frac{1}{2}\{\rho^a(p),\rho^b(q)\}$.

- **Coefficients** are real. They multiply $\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda}$, so the i of the three-ρ prefactor and the i of the commutator have been multiplied out.

There are 335 terms in total. Together with the genuine two-ρ terms of `Evolution_two_rho.tex` (with fixes F5, F6) and the two-ρ lists of `4rho.tex` (with fixes F1, F2), they give JIMWLK⊗LO exactly. This equation was re-read from this file and checked numerically against that sum.

$$\frac{d^3N_{\rm NLO}^{2\rho\,(3\rho)}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[$$

From four ρ's (d^3N_a, d^3N_b)

$$\begin{aligned}
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w)\,U^{ca}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ca}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^e(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^c(x')\,\rho^e(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^h(x')\,\rho^l(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{dh}(w')\,U^{hb}(w)\,\rho^a(x)\,\rho^e(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{bh}(w')\,U^{dh}(w)\,\rho^a(x)\,\rho^e(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^l(x)\,\rho^h(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^h(x')\,\rho^l(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)
\end{aligned}$$

Group I, Row II

$$-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$$

$$
\begin{aligned}
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ah}(w')\,\rho^{b}(x')\,\rho^{e}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,\rho^{b}(x')\,\rho^{e}(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+2\,N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ah}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ah}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^{l}(x)\,\rho^{h}(x')
\end{aligned}
$$

Group I, Row III

$$
\begin{aligned}
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')
\end{aligned}
$$

Group I, Row III: $\mathcal{D}$

$$
\begin{aligned}
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(w)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(w)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)
\end{aligned}
$$

$$
\begin{aligned}
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(w')\,\rho^l(x')\,\rho^h(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(x')\,\rho^l(x')\,\rho^h(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(w)\,\rho^h(x')\,\rho^l(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(w)\,\rho^l(x)\,\rho^h(x')\\
&- N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(w)\,\rho^l(x)\,\rho^h(x')\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{cl}(x)\,U^{be}(w')\,\rho^l(x)\,\rho^h(x')\\
&- N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x)\,U^{be}(w')\,\rho^l(x)\,\rho^h(x')\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x)\,U^{be}(x')\,\rho^l(x)\,\rho^h(x')
\end{aligned}
$$

Group I, Row IV

$$
\begin{aligned}
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&- \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&- \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ca}(x')\,\rho^c(x')\,\rho^b(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{da}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ae}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x)\,U^{cb}(x')\,\rho^c(x')\,\rho^a(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bd}(x')\,U^{lc}(x)\,\rho^h(x')\,\rho^l(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\ U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ U^{ab}(x)\,U^{ca}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{da}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+ N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ U^{ab}(x)\,U^{ca}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bl}(x)\,U^{dc}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+ N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')
\end{aligned}
$$

Group I, Row V

$$
\begin{aligned}
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{be}(z)\,U^{cl}(w')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bd}(z)\,U^{cl}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bl}(z)\,U^{cd}(w)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+ N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{bl}(z)\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+ N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')
\end{aligned}
$$

Group II, Row I

$$
\begin{aligned}
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ U^{ab}(w')\,U^{ca}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ f^{abc}\,f^{deh}\,U^{be}(z)\,U^{al}(w')\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{bm}(z)\,U^{md}(x)\,U^{ah}(w)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ U^{ab}(w')\,U^{bc}(w)\,\rho^{c}(x)\,\rho^{a}(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hm}(x)\,U^{ld}(x')\,\rho^{m}(x)\,\rho^{e}(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ce}(x')\,U^{al}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ U^{ab}(w')\,U^{ca}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{bm}(z)\,U^{md}(x)\,U^{ha}(w)\,\rho^{l}(x')\,\rho^{e}(x)
\end{aligned}
$$

$$
\begin{aligned}
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(w')\,U^{bc}(w)\,\rho^c(x)\,\rho^a(x')\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hm}(x)\,U^{ld}(x')\,\rho^m(x)\,\rho^e(x')\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{db}(z)\,U^{al}(w)\,\rho^l(x)\,\rho^h(x')\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ae}(w')\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{bl}(z)\,U^{lm}(x)\,U^{ah}(w)\,\rho^m(x)\,\rho^e(x')\\
&- N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{hd}(w)\,\rho^e(x')\,\rho^b(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hd}(x')\,U^{lm}(x)\,\rho^e(x')\,\rho^m(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ce}(x')\,U^{ad}(w)\,\rho^h(x')\,\rho^l(x)\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{bl}(z)\,U^{lm}(x)\,U^{ha}(w)\,\rho^m(x)\,\rho^e(x')\\
&+ N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x)\,U^{ca}(x')\,\rho^b(x)\,\rho^c(x')\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{hd}(w)\,\rho^e(x')\,\rho^b(x)\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hd}(x')\,U^{lm}(x)\,\rho^e(x')\,\rho^m(x)\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{lb}(z)\,U^{ad}(w)\,\rho^h(x')\,\rho^l(x)\\
&+ N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)
\end{aligned}
$$

Group III, Row I

$$
\begin{aligned}
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{md}(x)\,U^{mb}(z)\,U^{ah}(w)\,\rho^l(x')\,\rho^e(x)\\
&- \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hd}(x)\,U^{lm}(w')\,U^{al}(w)\,\rho^m(x')\,\rho^e(x)\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{mb}(z)\,U^{ah}(w')\,\rho^e(x')\,\rho^l(x)\\
&- \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hd}(x')\,U^{lm}(w)\,U^{la}(w')\,\rho^e(x')\,\rho^m(x)\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ae}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ld}(x')\,U^{me}(w')\,U^{am}(w)\,\rho^h(x')\,\rho^c(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{le}(x)\,U^{md}(w)\,U^{ma}(w')\,\rho^c(x')\,\rho^h(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{md}(x)\,U^{mb}(z)\,U^{ah}(w)\,\rho^l(x')\,\rho^e(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hd}(x)\,U^{lm}(w')\,U^{al}(w)\,\rho^m(x')\,\rho^e(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{le}(x)\,U^{md}(w)\,U^{lb}(z)\,U^{ma}(w')\,\rho^c(x')\,\rho^h(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ae}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ld}(x')\,U^{me}(w')\,U^{am}(w)\,\rho^h(x')\,\rho^c(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)
\end{aligned}
$$

$$
\begin{aligned}
&- \ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{mb}(z)\,U^{ah}(w')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+ \ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(w)\,U^{hb}(z)\,U^{la}(w')\,\rho^{e}(x')\,\rho^{m}(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\ U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+ N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w')\,U^{am}(w)\,\rho^{l}(x)\,\rho^{e}(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w')\,\rho^{m}(x)\,\rho^{e}(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w)\,U^{ma}(w')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+ \ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w)\,\rho^{m}(x)\,\rho^{e}(x')\\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&- \ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w')\,U^{am}(w)\,\rho^{l}(x)\,\rho^{e}(x')\\
&+ \ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w')\,\rho^{m}(x)\,\rho^{e}(x')\\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+ \ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(w)\,U^{hb}(z)\,U^{ma}(w')\,\rho^{l}(x)\,\rho^{e}(x')\\
&- \ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')
\end{aligned}
$$

Group III, Row II

$$
\begin{aligned}
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{lm}(x')\,U^{bh}(z)\,U^{al}(w)\,\rho^{m}(x')\,\rho^{e}(x)\\
&- \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&- \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ U^{ab}(x')\,U^{ca}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{bh}(z)\,U^{al}(w')\,\rho^{m}(x)\,\rho^{e}(x')\\
&- \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&- \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ U^{ab}(x)\,U^{ca}(w')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{be}(z)\,U^{ad}(w)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ah}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ae}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ah}(w')\,\rho^{c}(x')\,\rho^{e}(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+ \ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{lm}(x')\,U^{bh}(z)\,U^{la}(w)\,\rho^{m}(x')\,\rho^{e}(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ U^{ab}(x')\,U^{ca}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ \ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&- \ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ea}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ \ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ah}(w')\,\rho^{c}(x')\,\rho^{e}(x)\\
&- \ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+ \ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{be}(z)\,U^{da}(w)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\ U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&- \ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+ \ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\ f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&- \ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\ f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{bh}(z)\,U^{la}(w')\,\rho^{m}(x)\,\rho^{e}(x')
\end{aligned}
$$

$$
\begin{aligned}
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ca}(w')\,\rho^b(x)\,\rho^c(x')\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{am}(w)\,\rho^l(x)\,\rho^e(x')\\
&+ N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')\\
&- N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{am}(w')\,\rho^e(x')\,\rho^l(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,\rho^e(x')\,\rho^b(x)\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{cde}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w)\,\rho^l(x)\,\rho^e(x')\\
&- N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+ N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w')\,\rho^e(x')\,\rho^l(x)\\
&- N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,\rho^e(x')\,\rho^b(x)\\
&+ N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(z)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)
\end{aligned}
$$

Group III, Row III

$$
\begin{aligned}
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(x)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ae}(x')\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x')\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ad}(x')\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x')\\
&- N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x)\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)\\
&+\ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ae}(x')\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x)\\
&- N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)
\end{aligned}
$$

Group III, Row V

$$
\begin{aligned}
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(w')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{cd}(w)\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{cd}(w)\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{ce}(w')\,U^{bd}(x')\,\rho^h(x')\,\rho^l(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{ce}(w')\,U^{bd}(x')\,\rho^h(x')\,\rho^l(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(w)\,U^{bd}(x')\,\rho^h(x')\,\rho^l(x)
\end{aligned}
$$

$$
\begin{aligned}
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(w')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\\
&+ N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{ce}(w')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(w)\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\\
&+ N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cd}(w)\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\\
&\underline{\text{Group III, Row VI}}\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{eb}(z)\,\rho^l(x')\,\rho^h(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{cd}(x')\,\rho^h(x')\,\rho^l(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{eb}(z)\,\rho^h(x')\,\rho^l(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')\\
&+ N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&- N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^h(x')\,\rho^l(x)\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(x')\,\rho^h(x')\,\rho^l(x)\\
&+ N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{1}
$$